Conduction vs Insulation by RealQM
and the Missing Stiffness

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Abstract

Can RealQM, as a new alternative to textbook quantum mechanics, explain the difference between a conductor and an insulator? RealQM represents an N-electron system by N non-negative unit charge densities in ordinary three-dimensional space, one per electron, meeting at free boundaries and minimising the total energy — Coulomb interaction plus kinetic energy, with no other force and nothing fitted. It is a computable, parameter-free model at polynomial cost.

The difficulty is one of scale. Atomic energies are tens of electron-volts; semiconductor activation is 50–500 meV; metallic conduction sets in near 20 meV. Against the total energy of the system computed here those are 2×10^{-3}, 5×10^{-5} and 5×10^{-6} — so whether a material conducts is decided by a difference five or six orders of magnitude below the number a calculation returns, and must survive being extracted from it. Insulation is then not a separate phenomenon but the absence of that difference.

The answer depends on what the carrier is asked to be, and three arrangements give three regimes. On a chain of atoms with the valence electron sitting on its own kernel, the polarisability is extensive, $\alpha = 27.4N$ flat to 1% per atom, and the corrugation the partition must slide through is of order an electron-volt: an insulator, and outside the activated-hopping class this framework claims. On a wire — ion cores with the carriers standing off them, which is how matter is actually built — the corrugation falls to ≈240 meV, inside the 10–500 meV band those materials show. And when the coefficient of the kinetic energy is raised above the value $\frac{1}{2}$ that atomic spectra fix — a value an extended object has no obligation to share — it falls to ≈24 meV, thermal energy: the pinning no longer holds and the charge is free to move. Not as a metal, since non-overlap leaves no extended states to fill into bands, but as a charge lattice sliding collectively over the ions.

That last step identifies the obstruction as a missing stiffness. The framework's free interface admits a flat density at no cost, so $C = 0$ exactly — the defect of §6 — and the same absence leaves the penalty on a *modulated* density too weak to stop the carriers bunching into the lattice wells. One deficiency, two symptoms: a wrong equation of state and an insulator that should conduct.

What remains outside in every case is the rate. The carrier is a *free boundary*, in the relation a dislocation bears to slip, and a minimisation has no time in it: a conductivity needs a velocity variable and a dissipation channel, neither of which a static theory has.

1 The question, and the two formulations

Can RealQM explain the difference between a conductor and an insulator? That is the question of this article: whether the framework, given only the charges present and their Coulomb interaction, distinguishes the two — and if not, where exactly it stops.

The question is worth asking of a new framework because the established one answers it well, and it is worth being clear about how. Band theory takes a periodic potential, solves the one-electron problem in it, and fills the resulting bands: a partly filled band conducts, a filled band with a gap does not. That is a theorem plus a counting argument, it places sodium and diamond correctly, and it does so at the size of a crystal. Density functional theory then computes the potential self-consistently and predicts real materials at useful accuracy. What both supply rather than derive is the starting point — a periodic potential in the first case, an exchange–correlation functional in the second — and the many-electron Schrödinger equation from which they descend costs exponentially in the number of electrons, so it is not itself solved at the size where the question lives.

So the interesting question about RealQM is not whether it beats that, but whether a framework built on Coulomb alone, with nothing fitted and polynomial cost, reaches the same distinction by its own route. The two formulations should be put side by side before either is judged.

RealQM [1, 2] represents an N-electron system by N non-negative densities ψ_i, each carrying unit charge on its own domain Ω_i, the domains non-overlapping and their interfaces free to move to where the energy is least. The energy minimised is Coulomb's:

$$E[\{\psi_i\}, \{\Omega_i\}] = \sum_i \int_{\Omega_i} \Big(\tfrac{1}{2}|\nabla\psi_i|^2 - \sum_a \frac{Z_a}{|x - x_a|}\psi_i^2\Big)dx + \tfrac{1}{2}\sum_{i\neq j}\iint \frac{\psi_i^2(x)\psi_j^2(y)}{|x-y|}\,dx\,dy, \qquad (1)$$

subject to $\int_{\Omega_i}\psi_i^2 = 1$. There is no wavefunction in $3N$ dimensions, no antisymmetry and no self-interaction; the role played by the exclusion principle is played by the requirement that domains do not overlap. And — of particular relevance here — there is no smeared continuum anywhere in the construction: everything in it is a discrete unit charge occupying a region of ordinary space.

Minimising (1) gives stationarity conditions that take the form of Schrödinger's equation — N of them in three dimensions, coupled through the Coulomb potential and through the free boundaries — together with a condition fixing where each domain ends. This is not the $3N$-dimensional problem rewritten or restricted: the minimisation runs over the partition $\{\Omega_i\}$ as well as over the densities, and a condition fixing a domain's edge has no counterpart in a formulation that has no domains. The two share their physical content — kinetic energy and the Coulomb interaction, no other force and nothing fitted — and differ in what a solution *is*: one function on $3N$ dimensions, or N functions on three with free boundaries. Neither is a special case of the other.

The contrast is often compressed to "3 dimensions against $3N$", which is the most consequential difference but not the only one, and on its own it invites the reading that one equation has simply been rewritten in fewer variables. The physics entering is the same; four things differ:

	textbook quantum mechanics	RealQM
the physics	Coulomb interaction + kinetic energy — the same for both	
the solution	one function on $3N$ dimensions	N functions on three
the object	complex antisymmetric amplitude whose square is a probability	real non-negative density which *is* a charge density
the equation	one, linear, unitary	N coupled, nonlinear
the domains	none	free boundaries, found by the minimisation
the cost	exponential in N	polynomial in N
its reach	a molecule	a solid

The last two rows are what decides whether either can be brought to the question this article asks. The cost of the $3N$ equation grows exponentially in N, so its reach is a molecule and never a crystal: at the size where conduction is a meaningful question it is uncomputable, and only the second column is available at all.

Hartree and Kohn–Sham theory work with a drastic reduction of dimensionality, and so are no longer *ab initio*: what they compute is not settled by Coulomb's law and kinetic energy alone.

One further difference decides how the next section's argument applies here. The $3N$ equation is *linear* in Ψ and its evolution unitary; (1)'s stationarity conditions are *nonlinear*, since each density moves in the field of the others. That matters because the argument of §4 — that a closed system rings rather than conducts, so irreversibility must be imported — is an argument about linear unitary evolution. Nonlinearity is where irreversibility can arise without being inserted, which is what is claimed below for the friction a driven partition feels.

So when conduction is said not to be derivable from Schrödinger's equation, what is at issue is the linear $3N$ formulation and what can be extracted from it, not the equation itself, which appears here too. Whether the nonlinear form escapes the same argument is not settled here for a simpler reason: as used below it is a *minimisation*, with no time in it at all, so it cannot ring or conduct or dissipate. That is a statement about this article's method, not about the framework's prospects.

A two-component system of ions and electrons is therefore not an idealisation the theory must be adapted to describe. It is what the theory is already made of, in a new arrangement.

Every previous test of this construction — ionisation energies across the periodic table, molecular binding, nuclear binding by dual Coulomb confinement [3] — has been on *bound* matter, with an attractor always present. A plasma is the first case in which large numbers of charges of both signs are free to arrange themselves, and it is a fair question what the framework then gives.

2 Which of them can answer it

What the $3N$ equation itself can contribute is limited, and the limit is worth stating precisely, because it is not a failure of physics but of arithmetic. Kohn established that the conductor/insulator distinction follows from the ground state alone, with no transport theory required; what that does is convert the question into a second one, produce the ground state. Solving the many-electron Schrödinger equation for that ground state costs **exponentially** in the number of electrons, so its reach is a molecule and never a crystal, and no growth in computing changes an exponent. Nor does the question live where the equation can be solved: whether a response is bounded or unbounded is a statement about how it grows as the system does — an isolated finite sample polarises and stops whatever it is made of — so the distinction exists only for an extended system. The systems the exact equation can reach are not ones in which the question can be posed, and the systems in which it can be posed are not ones it can reach. The two domains do not overlap. A criterion whose input can never be supplied, for any system to which the criterion applies, is not an answer held in reserve; it is a description of what an answer would look like.

What is used at the size of a solid therefore answers a different question. Band theory supplies the periodic potential instead of computing it and counts fillings; it places sodium and diamond correctly and NiO wrongly, a partly filled band there belonging to an excellent insulator. Density functional theory supplies an exchange–correlation functional. A Hubbard treatment supplies U and t. Each returns a verdict about a model that was put in, and the distinction that was to be explained is among its inputs.

The second formulation is not blocked in the same way. RealQM costs **polynomially** in the number of electrons and carries no fitted quantity, so the question can be put to it at the size at which it is meaningful, and put from scratch. That is the whole of its claim at this point: not that it answers correctly, but that it is the one of the two that can be asked.

The rest of the article asks it. Sections 4 and 5 fix what would count as an answer and what kind of conductor this construction could be; the remainder puts the question to it and reports what comes back — which is a clean verdict on one half and an obstacle on the other.

The whole of the experiment, said in advance. It is two measurements on one chain of seven model atoms carrying one valence electron each, and the reader is owed that before the argument that surrounds them. *First:* put the chain in a uniform field, hold the partition fixed, and read the induced dipole — repeated at three chain lengths, this gives α and how it scales. *Second:* apply no field at all; displace the whole partition rigidly by u and record the energy — this gives the periodic curve $E(u)$, whose amplitude is the corrugation and whose steepest slope is the field at which the pattern would begin to run. Everything else is control: the same converged states reassembled under three boundary conditions, a range of spacings and fillings, a mesh ladder, and the requirement that a shift by one full spacing return the same energy. The article is long because the arguments are, not because the apparatus is.

3 The extension, and the object it is made for

In this article we extend RealQM, developed for atoms and molecules, to **two-component systems**: ions and electrons, both charged, both free to move, interacting by Coulomb forces and nothing else. A metal is such a system, and the question pursued here is the elementary one such a system poses — whether it *conducts*, and if so what carries the charge. The same construction covers a stellar interior and the recombination era of a Coulomb cosmology, treated as a classical plasma.

There the one exact result is the **plasma frequency**, and it is worth saying what it is. Displace the whole electron population rigidly by u against its neutralising ion background: charge neu per unit area is exposed at the two faces, the field between them is $4\pi neu$, and the restoring force per electron is $-4\pi ne^2u$. The population therefore oscillates at $\omega_p^2 = 4\pi ne^2/m$, or $4\pi n$ in atomic units, and the construction returns it exactly.

That is a check on the Coulomb bookkeeping of a two-component system, not evidence for the quantum structure: the derivation is electrostatics and Newton's law, and any theory with the same electrostatics returns the same number. It is quoted here to establish that the extension to ions and electrons is set up correctly before anything harder is asked of it. The quantities that *would* test the quantum structure — the screening length and the dispersion of that oscillation — are where the construction is found wanting, in §6.

4 What conduction asks of a theory

Before asking whether this construction conducts it is worth being exact about what the question demands, because the obvious standard is one that no framework meets.

Conduction is not derived from Schrödinger's equation. The equation is unitary and reversible while resistance is dissipative, and a closed quantum system in a perfect periodic potential, placed in a constant field, reaches no steady current at all: k runs to the zone boundary, Bragg-reflects and returns, so the system rings — *Bloch oscillations* — rather than conducts. A steady current requires irreversibility, and every treatment imports it rather than deriving it: a collision term in Boltzmann transport, a partial trace in an open-system treatment, thermalising reservoirs in Landauer's, τ as an input in $\sigma = ne^2\tau/m$.

What it does supply is the states. Bloch's theorem gives extended solutions in a periodic potential, and Pauli filling then gives the metal–insulator *classification* directly: a partly filled band conducts, a filled band with a gap does not. That is the standing of the question below — whether this framework possesses states of the kind that fill.

So the honest question comes in two tiers, and only the first is a fair test of a structural theory:

tier	status
the classification: which materials conduct	from bands and filling; *this is the fair bar*
the value of σ	needs added statistical machinery in every framework

This article asks the first question of RealQM. It does not ask the second, and no comparison here should be read as faulting the construction for failing to supply a conductivity, since nothing supplies one unaided. What it does examine is whether the framework can reach the classification at all — whether it possesses states of the kind that fill into bands — and the answer, developed below, is that it does not, for reasons that are structural and that the framework's own postulate makes explicit.

Which clock is missing. The first tier needs no time — that is Kohn's point, and it is what lets a static calculation classify. The second does, and the objection to be met is that a minimisation could hardly report a current whatever it was applied to, copper included. So it is worth being exact about which time is absent rather than calling the framework static. A wire contains three time scales, twenty orders of magnitude apart, and the construction lacks them for three different reasons:

what moves	its scale	why this framework does not have it
the signal, hence the energy	$\sim c$	(1) is instantaneous: no retardation, no Maxwell
the carriers	mm/s drift	this *is* σ: inserting it assumes the answer
the electron state	$\hbar/E_h \approx 24$ as	a real ψ has no phase to evolve

Only the last is intrinsic — c arrives by coupling to Maxwell and drift is a transport coefficient — but restoring it does not by itself produce a conductivity. Attosecond variation is far below the resolution of any conduction measurement: scattering times are femtoseconds to picoseconds and a conductivity is an average over many of them, so the phase evolution is integrated out before an ammeter sees anything. What it supplies is not the rate but the *object* — a current, $J \propto \Im(\psi^* \nabla \psi)$, which is identically zero for a real density and cannot be made nonzero by any refinement of a minimisation. What is absent is a *velocity* variable, and that is a weaker requirement than a complex amplitude: in Madelung's hydrodynamic form [4] the same content is carried by a real density together with a real velocity field, with $J = \rho v$ and a continuity equation in place of the phase. The vanishing of $\Im(\psi^* \nabla \psi)$ is therefore a statement about a minimisation, which has no velocity in it at all, and not about real densities, which can perfectly well flow. The rate still comes from the dissipation of the table above.

These two absences are one. A real non-negative density has no phase, and the same lack makes the theory a minimisation rather than an evolution *and* makes the current within a domain vanish identically. That has a positive consequence, and it is the one this article follows: if charge cannot flow inside a domain, the only object left that can carry it is the boundary between domains. It also fixes how the question must be put. A framework with a clock answers "does it conduct" by driving the system and reading a current; this one cannot, so the question is recast as a static one — the maximum gradient of $E(u)$, and the scaling of α with N — which is what §8 and §9 do. The missing clock does not supply the verdict; it determines the form in which a verdict is available at all.

And a pseudo-time is not a clock. There is a time parameter in the solver, and it is worth saying why it is not this one. The fields relax in *imaginary* time, $\tau = it$, under which $e^{-H\tau}$ damps every excited component as $e^{-(E_n - E_0)\tau}$: a filter that finds the ground state, not a dynamics that evolves a state, and the oscillation is not lost by accident but removed on purpose. Charge conservation goes with it. In real time $\partial_t \rho + \nabla \cdot J = 0$ holds exactly; continued to imaginary time the same equation becomes a diffusion with a source, with no flux to conserve and the norm restored by a renormalisation that is global. The relaxation is therefore not an underpowered dynamics that a smaller step would repair — it is the branch of the equation from which the current has been removed.

What does conserve charge locally, in the scheme as it stands, is the interface rule: cells change ownership between neighbouring domains by the local density balance $(\rho_i - \rho_j)/(\rho_i + \rho_j)$, so what leaves one domain arrives in the next. That is a genuine local flux, and it is the reason the boundary and not the electron is the carrier here. It is also where a time-dependent formulation should begin — a continuity law with the boundary velocity as the transport variable — rather than in promoting a descent that was never a conservation law. The descent itself can be promoted in exactly two ways, neither of which answers the question. Overdamped, $\partial_t\Omega = -M\,\delta E/\delta\Omega$, gives a boundary velocity proportional to the driving force — Ohm's law with the correct structure, except that the mobility M is put in by hand and is the conductivity up to geometry. Inertial, giving the moving interface a mass, gives oscillation: the chain rings, as any closed system does, and carries no steady current. The descent rate is a numerical parameter with no calibration to anything physical, and each way of calibrating it either inserts the answer or returns the ringing.

5 What kind of conductor this can be

The construction does not have a free choice of target, and identifying it correctly settles most of what follows. Non-overlap fixes two things at once, and they happen to be the two parameters of the Hubbard Hamiltonian. The on-site repulsion U — what it costs to put two electrons on one site — is infinite, since two electrons may not share a domain at all; and the transfer integral t — the amplitude to hop to a neighbour — is exactly zero, since it *is* the overlap of neighbouring orbitals and there is none. In that language the construction is the **atomic limit** and not the Mott limit — the reading it is usually given — and the difference is not a refinement:

	$U \to \infty$, $t \neq 0$	$U \to \infty$, $t = 0$
half filling	Mott insulator	isolated atoms
superexchange	$J = 4t^2/U$	none
doped	**conducts**	still nothing
spin dynamics	yes	none

A Mott insulator is insulating but alive: it carries magnetic exchange, and doping it produces a conductor, which is what the cuprates are. The atomic limit is a collection of atoms with electrostatics between them. This accounts for every negative result below, and for one in particular: doping a Mott insulator works *through* t, so at $t = 0$ there is nothing for doping to activate — which is exactly what the measurement shows, the barrier rising with carrier concentration rather than falling. Metals are therefore out of scope in kind and not by degree, their carriers being extended states and their transport band motion, neither of which an arrangement of densities on domains appears able to supply. Two results usually read as failures follow from that scope rather than against it: $d\sigma/dT > 0$ is *correct* for an activated conductor and fatal only for a metal, and the absence of a Fermi surface does not bear on a dilute carrier gas, which is non-degenerate and has none to reproduce.

What remains within reach is matter whose carrier is a localised charge on a site, whose gap is on-site repulsion rather than band structure, and whose transport is activated hopping. That is a real class — doped Mott insulators, organic and molecular semiconductors, polaronic oxides, amorphous semiconductors — and this article set out to claim it. §7 measures the activation energy and finds 613–1723 meV, against the 10–500 meV those materials show. The claim does not survive its own measurement, and what the construction describes is the insulating end of the class and not the conducting one. Silicon and the III–Vs were never in scope, their gaps being band gaps.

6 The interface, and the one number that is wrong

The free interface has a defect of its own, recorded here because it conditions everything the partition does, though the hopping barrier does not turn on it. Partitioning a gas of density n gives a kinetic energy density $t(n) = C\,n^{5/3}$ — the Thomas–Fermi *form*, obtained from partition geometry rather than Fermi statistics — but the flat trial function is admissible on a free boundary, so $C = 0$ exactly. There is no degeneracy pressure, the screening length vanishes rather than being finite, and the plasmon does not disperse. The defect is one number and not the construction: C is fixed by the interface condition, and a Robin condition at $\beta a = 1.146$ gives $C = 2.871 = C_{\mathrm{TF}}$ exactly, against Dirichlet's 14.804, all three carrying the correct $n^{5/3}$ scaling.

Whether that value is tolerable for matter is the open question, and it is sharper than a parameter check. Adding the surface term and scanning it, H_2 unbinds by $\beta a \approx 0.71$, below the 1.146 the electron gas requires: Teller's theorem, that Thomas–Fermi binds no molecule, is met inside the framework rather than inherited from outside it. Double occupancy — the repair conduction would need at half filling — becomes favourable at $\beta a = 0.451$ and is affordable by comparison. These figures were obtained at differing densities and are not strictly commensurable; the equation of state is pursued elsewhere, and what matters here is only that the interface carries no cost of its own unless one is supplied.

A caution about specified partitions, learned by getting it wrong. The method of §7 evaluates the energy of a *stated* partition, and it is sound only when the stated partition is one the physics admits. The interface is not a surface to be shaped at will: the wavefunction is a single global field partitioned by labels, continuous across every interface, each domain carrying one unit of charge, and the boundary sits where the densities balance. Freezing a partition removes that last condition. An attempt to measure a bending rigidity by imposing a curvature, freezing it and relaxing the fields gave an energy falling symmetrically and quadratically in the imposed curvature, which reads convincingly as a negative rigidity; but released, a boundary started from the curved configuration returns to the flat one. The flat interface is the balance surface and is stable, and what the frozen scan measured was the energy gained by *dropping the balance condition.* A specified partition gives a legitimate energy difference between configurations the physics admits — as symmetry guarantees for the two in §7 — and a meaningless one along a direction it forbids. Nothing in the numbers distinguishes the two cases; the check is to release the boundary and see whether it stays.

7 Computing with it anyway

Electric conduction is the interaction of the electrons with the ion lattice. It is the elementary case — ordinary matter at ordinary temperature — and it is for that reason the *demanding* one here, since ordinary-temperature conduction is degenerate matter, and degeneracy is what this framework has least of.

One property of the construction is needed before anything else, and it is the property that makes conduction a live question rather than a hopeless one. With the free interface used throughout, a uniform density is admissible on any partition: $\psi_i \equiv |\Omega_i|^{-1/2}$ satisfies the normalisation, minimises the electrostatic energy against a neutralising background, and has $\nabla\psi_i = 0$. Redrawing the boundary therefore costs nothing,

$$\text{a free interface between domains of equal density carries no confinement energy,} \qquad (2)$$

and the boundary is the only object this framework moves — its solver evolves each interface by the local density balance $(\rho_i - \rho_j)/(\rho_i + \rho_j)$, transferring ownership of a cell when the neighbour's density exceeds its own. Domains do not translate; they are redrawn. *A vanishing interface cost is a vanishing cost of transport.* What that same fact does to the equation of state is a separate matter and is taken up in §6, where it is much less welcome.

Metallic conduction is the paradigm of *delocalisation*: the carriers are extended states spanning the crystal, whereas RealQM's electrons are localised by construction, each in its own domain, with non-overlap requiring N carriers to occupy N disjoint regions. The natural minimiser of (1) is a set of compact cells, which describes a Mott insulator more naturally than a conductor. Nor is there an evident mechanism for the metal–insulator distinction, which in standard theory is band filling — two electrons per band per cell — an exclusion-principle count the framework does not have; and with no Fermi surface, the carrier density in $\sigma = ne^2\tau/m$ has no referent.

What does fit naturally is the opposite regime: localised charges hopping between domains by thermally activated jumps, as in semiconductors, disordered systems and small-polaron transport. That needs no Fermi surface and follows from the same thermal motion that supplies the pressure.

This yields a prediction crude enough to be decisive, and it concerns only a sign. Metallic conductivity *falls* with temperature; hopping conductivity *rises*. If RealQM transports charge by hopping rather than band motion it should give $d\sigma/dT > 0$. Within the scope set in §5 that is the *right* answer, activated transport being what a localised-carrier semiconductor does; it would be fatal only for a metal, which is why the target matters. Whether it does has not been computed for electrons, and the pessimistic reading should be held loosely: the same lattice that reproduces alkali lattice constants is present here too.

For *protons* the mechanism has already been demonstrated in this framework. Grotthuss conduction — an excess proton migrating along a hydrogen-bonded chain of water molecules by successive transfers, rather than by bodily motion of any one particle — is hopping transport in its purest form, and it is a standard RealQM calculation: a chain of waters with O–O spacing near 2.65 Å, an excess proton, and the question whether it hops from one molecule to the next. Related transfers, $HF + H_2O \to F^- + H_3O^+$ and $HCl + NH_3 \to Cl^- + NH_4^+$, are computed the same way.

This matters for the argument rather than merely illustrating it. Charge transport by transfer between localised domains is not a regime the framework must be stretched to reach — it is one it already handles, in the case where the carrier is a proton.

The electron case cannot presently be tested in the same way, and the obstruction is instructive enough to be worth setting out. Two chain calculations were attempted here — five hydrogens with an excess electron, and five hydrogen kernels with one electron missing, the electronic counterparts of the proton wire — and both returned nothing. Neither result means what it appears to.

The cause is not what it first appeared. Kernel proximity assigns the labels only *initially*; thereafter a free-boundary rule evolves each interface from the local density balance and transfers ownership of a cell when the neighbouring domain's density exceeds its own. Partitions therefore do rearrange, and electron transfer is representable in principle. What defeated the attempts here was specific to each: a carrier given no kernel feels no force from the lattice; a bare kernel is given no domain, so a vacancy has nothing to migrate into; and a carrier placed on an occupied site can seize the nuclear region outright, since with free interfaces no confinement cost opposes it. Each is a property of the construction, not of (1).

The deeper cause survives that repair. Conduction is charge in motion, and (1) is an energy minimisation: it returns the equilibrium partition, not a current. Given complete freedom in the labels one would still obtain the ground state, and for a symmetric chain that state places one domain per site by symmetry, with nothing flowing. A minimisation has no time in it.

Which filling the measurements belong to, and which needs the repair. It is worth being exact about this, because it determines how much of the article depends on β. At *half filling* — one electron per site, every domain occupied — the $U \to \infty$ construction forbids conduction outright: there is no vacancy for a carrier to enter, and the insulator is structural rather than dynamical. That is the case the double-occupancy repair is for. But it is not the case measured here. The chains reported above are doped by ionising every second atom, leaving half the sites empty, and every barrier in this article is the cost of a carrier moving into a *vacancy*. Single occupancy handles that without amendment.

So the transport results stand on their own: they require no finite U, no double occupancy and no β. What requires the repair is the undoped, half-filled chain, which is not run here. This is the stronger position of the two — the measurements are independent of the proposal made to extend them — and it locates precisely what the proposal would buy: not better hopping, but conduction at a filling where the construction currently permits none.

What the framework does possess is an asymmetry of dynamics. Nuclei move, under classical equations of motion; electrons relax, to a static minimum. Charge carried by *nuclei* is therefore representable — which is precisely why the Grotthuss calculation works, the carrier there being a proton — and charge carried by electrons is not, because in this scheme electrons have no equation of motion at all. The correct machinery exists elsewhere in the programme, in the time-dependent form where a real density carries a complex current, and a conduction calculation belongs there rather than here.

8 The threshold field, measured

The question this article asks has a form that avoids barriers altogether, and it is the form in which the answer can actually be obtained.

Put the chain in a field. Every electron shifts to the right. If the shift reaches one full lattice spacing, each electron has taken over its neighbour's kernel — and since the electrons are identical and the energy depends on the partition and not on labels, *the configuration is restored.* That restoration is exactly what separates a current from a polarisation:

tilt below the steepest slope of $E(u)$	u saturates: bounded polarisation, no DC current
tilt above it	u runs: the state cycles, charge flows steadily

This *is* the applied-field, free-boundary case, and it is worth saying so plainly: u is the collective coordinate of the free boundaries themselves, the displacement a field would drive, and a uniform field adds nothing to the energy but a term linear in it, $E(u) - NeFu$. The field therefore tilts the landscape and changes nothing else in it, so the landscape can be computed once, without a field, and tilted afterwards. What is given up by doing it this way is the motion *after* the last stationary point is lost — the rate, which the paragraph below shows would be an artifact in any case — and not the threshold, which is exactly where that point is lost.

So the threshold is not an activation energy and not a saddle. It is the **maximum gradient** of the periodic energy $E(u)$, and §11 has measured that curve on the wrapped chain, where $E(u+a) = E(u)$ holds to every digit and the three reflection pairs agree to 1–2 $\times$ 10^{-5} Ha. Nothing here differences distant configurations.

From the measured curve at $r_c = 1.93$, $a = 5.5$ a_0 — both parameters determined rather than chosen — the steepest segment gives $|dE/du| = 0.0982$ $\mathrm{Ha/a_0}$ for seven electrons, and a sinusoidal fit, which recovers the peak a chord underestimates, gives 0.1194. Per electron:

$$F_{\text{dep}} = 0.014\text{–}0.017 \text{ au} = 7\text{–}9 \times 10^9 \text{ V/m} = 2.1\text{–}2.6 \text{ V per lattice site.} \tag{3}$$

Copper carrying an ampere runs at 4×10^{-12} V per site. The separation is **twelve orders of magnitude**, and it is not a number any refinement moves. Nor is such a field available: dielectric breakdown occurs two to three orders of magnitude below 10^{10} V/m, so the lattice is destroyed long before the electron distribution depins. *The construction does not conduct under any field a material survives.*

Above the threshold, and why the rate there is not a result. Drive harder than (3) and the partition does run: the tilt exceeds every slope of $E(u)$, no configuration is stationary, and the solver will advance the interfaces without stopping. It will also report a speed, and that speed is worth naming as an artifact before anyone quotes it. What sets it is the descent parameter of the relaxation scheme, not the physics — as §4 sets out, the pseudo-time carries no calibration — so the same chain under the same field yields whatever current the numerical mobility is set to. The threshold is a property of the measured curve $E(u)$ and survives refinement; the current above it is a property of the scheme and does not — and the regime is in any case the one just excluded.

Why this supersedes the barrier attempts. Every earlier route asked for a difference between two configurations far apart in the energy landscape — and §11 shows why that could not work, six of eight scans failing their own control. Equation (3) asks instead for the slope of a single curve, sampled at points a fraction of a spacing apart, on a geometry whose control passes by four orders of magnitude. It is the same physics, asked in the form the discretisation can answer.

A defect in the model behind these numbers, and what it leaves standing. The pseudopotential exclusion used throughout this section keeps valence density out of a domain's *own* core only. A cell belonging to domain A but lying inside neighbouring core B is not excluded, so A's density enters B's core and collects its attraction. That is invisible wherever each domain contains its own core — which is why $d = 0$ is unaffected — and decisive at $d = \frac{1}{2}$, where every core is split between two domains and the exclusion never fires on either half. Excluding from *every* core instead inverts the registry: the minimum moves to $d = 0$, atom-centred, and the bond-centred preference reported here disappears.

Neither treatment is usable as it stands. Own-core exclusion under-excludes and gives the wrong registry; full hard exclusion over-confines, since at $r_c = 1.93$ and $a = 5.5$ the excluded spheres leave only $1.64\,a_0$ between neighbours and the electron at $d = \frac{1}{2}$ becomes a particle in a box rather than a charge in a lattice. The two therefore *bracket* the answer rather than settling it, and what is needed is a soft repulsive core, or r_c recalibrated with full exclusion active. $V_0 = 816 \pm 4$ **meV and the threshold field** (3) **are the own-core values**, verified as such but resting on an exclusion rule that is not the model. Refinement cannot repair this: the mesh convergence below is convergence of the wrong quantity.

The scaling route of §9 does not inherit the defect. Its measurements are made at an undisplaced partition, where each domain contains its own core and the two rules agree to five decimals, so $\alpha = 27.4N$ and the extensivity verdict stand independently of how the exclusion is settled. That the article reaches the same conclusion by two routes is what makes the loss of one of them survivable.

And it is converged. The amplitude was checked on three grids at fixed geometry, $a = 5.5$, $r_c = 1.93$, with the translation control exact and the reflection pairs identical to five decimals at each:

h (a_0)	0.448	0.352	0.299
V_0 per electron (meV)	812	820	815

$V_0 = 816 \pm 4$ meV, a spread of half a per cent across a fifty per cent refinement. What makes that possible is worth stating, because the same solver produced scatter of a factor of two in §11: over this range the *totals* drift by 1572 meV while the amplitude moves by 7. The cancellation is 99.5%, against roughly 89% for the open-chain barrier — and 89% left a residual comparable to the signal, where 99.5% leaves one per cent of it. The improvement comes from the geometry alone: $d = 0$ and $d = \frac{1}{2}$ lie on one periodic curve on an identical grid with identical topology, so their discretisation errors track each other, whereas the open chain's two configurations differed in their end domains as well as their registry and the errors were independent. *A difference is only as good as the symmetry relating the two configurations*, which is the transferable lesson of this article.

The corrugation is bounded below, at every density. The same scan repeated across lattice spacings, with the translation control exact (0.0 meV) at every one and reflection pairs agreeing to 2–3 $\times 10^{-5}$ Ha:

a (a_0)	3.5	4.5	5.5	6.5	8.0
V_0 per electron (meV)	1513	972	812	**783**	816

V_0 has a *minimum* near the equilibrium spacing and rises on both sides: compressed, the cores crowd the electron; expanded, sharing a kernel at the bond-centred registry costs more than owning one. So the corrugation is bounded below at ≈783 meV per electron across a factor of 2.3 in lattice constant, never approaches zero, and the registry preference never flips — there is **no free-sliding point at any density**.

That settles a question the structural argument leaves open. A framework that could reach $V_0 \to 0$ would produce both phases and could decide conduction on its own terms; this one produces only the pinned phase. And the trend runs the wrong way: real matter *metallises* under pressure, as screening flattens the corrugation, whereas here compression makes the pinning worse. This also restores the compression result reported earlier by the free-boundary route which §11 discredits — a ratio between configurations sharing partition family and mesh is far more robust than V_0's absolute value.

What is bounded, and what is not. Everything measured here is the *rigid* slide: all N electrons displaced together, at half filling, where the conducting electrons are also the binding electrons. That is the expensive path, and its cost grows with chain length. The cheap path is a single carrier moving while the rest hold still — a kink, whose cost is length-independent — and that is the configuration real conductors use, a supporting lattice that binds and a dilute carrier that conducts. *The carrier barrier is not measured in this article.* A kink on the cyclic geometry requires the slip to accumulate to exactly one unit around the loop, $f(j+N) = f(j) + 1$, which the profile used here does not satisfy; the attempt fails its own control by 997 meV and is not reported as a result. Note also that the maximally delocalised limit of such a profile gives evenly spaced walls and a barrier of exactly zero, so the kink barrier is a function of its localisation and the physical value is the minimum over that — a two-parameter determination, not a single number. **The figures above are therefore upper bounds on the quantity conduction depends on**, and the verdict rests on them together with the structural argument, not on a measured carrier barrier.

A third qualification, and the largest. Every energy in this article is computed with the **nuclei clamped** at their lattice sites. So $V_0 = 812$ meV is the rigid-lattice corrugation, and (3) the rigid-lattice threshold — both upper bounds, because in real matter the lattice relaxes around the carrier and that relaxation is the dominant term. It is what makes small polarons work at all:

the Marcus–Holstein barrier is $E_a = \lambda/4$ with λ the reorganisation energy, and $\lambda \approx 0.1$–0.3 eV in organic semiconductors gives 25–75 meV, far below the bare electronic corrugation of those same materials. Nothing here tests it, and the framework is not prevented from doing so — nuclei move in this scheme under classical equations of motion, which is the half of the dynamics it possesses and the reason the Grotthuss calculation works. If lattice relaxation reduced the corrugation by the factor it does in polaronic materials, 812 meV would fall toward 100 meV and the threshold with it, from $\sim$8 to $\sim$1 GV/m: still above dielectric breakdown, but no longer by three orders. This is the one accessible variation that could move the verdict rather than confirm it, and it is untested.

The two bounds are independent and they multiply, which is worth stating in one place rather than leaving to be assembled: a clamped lattice and a rigid slide both raise the number reported, the first by an estimated order of magnitude and the second by a factor nobody here can quote, since the kink barrier is the quantity §11 shows cannot be extracted. Twelve orders is therefore the separation for the mode measured, not a floor under every mode. What survives either way is the sign of the conclusion — the field required is far above the one at which a wire carries current, and the reduction would have to reach nine orders to change it.

Two qualifications, both real. This is *rigid* sliding, every electron moving at once, which is the expensive path — the electron spacing is held fixed, the whole pattern translating without deforming — so a kink, which compresses the domains ahead of it and expands those behind, depins at a lower field and (3) is an upper bound. And the chain is unstable to non-uniform distortion — a sinusoidal displacement *lowers* the energy, giving an effective stiffness $\kappa < 0$, because domains here cannot change position without changing width and a wider domain lets its electron spread. What the distorted state does under a field is not determined here. Both qualifications point the same way, toward a lower threshold, and neither is remotely capable of closing twelve orders of magnitude.

Why a chain rather than a ring. A ring supplies a clean barrier — no ends, every site equivalent, the energy obliged to repeat with the lattice period — but it cannot host the drive: a conservative field has zero circulation, so no bias drives a sea around a closed loop. A chain has ends, a potential drop along it is an ordinary thing to apply, and a single carrier may enter at one end and leave at the other, which is the transport event itself.

The partition must be specified, not searched for. Before the number, the method, because the obvious approach fails and the failure is instructive. A kink is *not* the ground state, so a free-boundary minimisation relaxes it away — asked to hold one, the solver oscillates between the kinked and uniform partitions and the energy never settles, wandering over 0.05 Ha against a barrier of order 0.01. That is not a convergence failure to be tuned away: it is unaltered by the surface tension β, by reducing the boundary step fivefold, and by freezing the labels outright.

The cause is general and it disqualifies the free boundary for every transport quantity in this article. The interface moves by *local single-cell label flips*, so a run descends to whichever configuration no single flip improves — the nearest stuck point, not the minimum — and which one that is depends on the initial condition. Four independent geometries agree:

geometry	spread	against a barrier of
ring, between two builds	0.005 Ha	0.012–0.022
kink chain, across steps	0.05 Ha (never settles)	~ 0.01
$H^- + H$, four seeds	0.031 Ha	~ 0.001
alkali chain, two seeds	0.013 Ha	~ 0.001

In every case the ambiguity exceeds the quantity sought, by one to two orders. It is a failure of the *search*, not of the energy functional, which is why nothing in the functional touches it. It is also why the ring's 0.01186 Ha moved to 0.022 Ha under a recompilation that altered no physics: rounding at the last bit flips cells sitting near the threshold, and a different stuck point follows. The rigid-sliding figure should be read as a bound of that order rather than a determination, and the doping barriers of the earlier barrier scans, obtained the same way, with it.

The repair is not a better minimiser but no minimiser. The domain walls of a chain are planes perpendicular to the axis, so they can be *placed*: specify the partition, freeze it, and relax only the wavefunctions. Nothing is searched for, the same input returns the same output, and the question asked is no longer "where does the boundary go" but "what does this configuration cost" — which is the question a barrier actually poses. It is the device that makes the H_2 midplane exact, applied to a coordinate instead of a symmetry.

9 The same verdict from scaling

The measurements in this section are on a *different chain* from §8, and that should be said before any of its numbers appear. It is a model atom of nuclear charge +1 with core radius $r_c = 1.0\,a_0$ and one valence electron, seven of them in a row at spacings from 3.5 to $8.0\,a_0$ — parameters chosen to define a system rather than to represent an element. The row is an arrangement of kernels, not a property of the construction: the domains are regions of ordinary three-dimensional space, and the same computation runs unchanged on a square or cubic lattice of the same atoms. Its corrugation runs from 182 to 461 meV per electron across that range, and is 379 meV at $a = 4.5\,a_0$, the spacing at which the polarisabilities below are computed. Nothing here is a second determination of the quantities of §8; what is claimed is a statement about *scaling*, which is independent of which chain it is made on.

That independence is worth making concrete, since the natural objection is that the verdict rests on two chains rather than one. Extensivity does not involve r_c at all: (5) is proportional to N for *any* positive stiffness, so r_c can move α_{intra} and K — both per-atom constants — without touching the power of N. What r_c could do is take the one escape, $K \to 0$, and the chain of §8 is measured to be further from it than this one: $V_0 = 812$ against 379 meV per electron, hence $K \simeq 530$ against 369 meV/a_0^2 and a slide term $e^2/K \simeq 51$ against $73.6\,a_0^3$. The stiffer chain is the safer one. Recomputing α at $r_c = 1.93$ would therefore move the coefficient 27.4, which is reported as a measurement of this chain and not as a property of the framework, and would leave the classification where it stands.

The threshold field settles the question by driving the pattern. It can also be settled without driving anything, by asking how the response grows with the size of the sample — an independent route, resting on different assumptions, and worth having because the two can be checked against each other.

One point about what is being asked deserves stating before any of it, because it decides which part of the measurement carries the weight. An isolated finite system cannot conduct, whatever it is made of. Place a piece of copper in a box and apply a static field: the charge redistributes to the surfaces, the interior field cancels, and everything stops. Copper polarises and stops. So the fact that a response *saturates* is never evidence of insulation — it is what any closed system does, and no calculation on a bounded sample can conclude otherwise.

What distinguishes the two cases is not whether the response saturates but how the saturated value *scales*. A metal's polarisability grows faster than the number of atoms; an insulator's is extensive. That is what Kohn's argument supplies and what makes a closed, static calculation capable of answering the question at all. The measurement that carries the classification here is therefore $\alpha = 27.4N$ — the extensivity, not the boundedness — and the corrugation enters only because a vanishing K is the one way extensivity could fail.

9.1 Why the scaling is fixed before it is measured

Nor is the outcome fixed in advance by the postulate. At full filling the domains tile space, but transport does not require a vacancy: every boundary may shift at once, translating the whole partition. That collective mode has a polarisability. Writing K for the per-electron curvature of the corrugation the partition slides through,

$$E(u) = \tfrac{1}{2}K_{\text{tot}}u^2 - NeFu \quad \Longrightarrow \quad \alpha_{\text{slide}} = \frac{N^2e^2}{K_{\text{tot}}} = \frac{Ne^2}{K}, \tag{4}$$

since the stiffness is extensive, $K_{\text{tot}} = NK$.

Each domain carries exactly one electron. The constraint $\int_{\Omega_i} \psi_i^2 = 1$ is imposed, not obtained, so the monopole of every domain is pinned and **no charge passes between sites**. The sites, however, are not the fixed points of the problem — the kernels are. A domain may arrive at the kernel that was its neighbour's while carrying the same unit of charge it began with, so transport here is *the sites moving*, not charge moving between them, which is the dislocation relation again and the reason nothing needs to hop. That removes the mechanism which gives extended systems their large polarisability, and leaves exactly two responses to a field: the density deforming within each fixed domain, and the domain boundaries displacing. Write α_{intra} for the first — the polarisability an atom would have with its domain held rigid, the electron leaning toward the field while its walls stay put — and use (4) for the second:

$$\alpha = N\Big(\alpha_{\text{intra}} + \frac{e^2}{K}\Big), \tag{5}$$

which is proportional to N exactly. Allowing the walls to move differentially does not change this: with elastic coupling κ between walls and substrate stiffness K per site, the displacement profile satisfies $\kappa u'' - Ku = -f$ and returns $u \simeq f/K$ through the bulk with corrections over a length $\sqrt{\kappa/K}$ at the ends, so the total dipole remains proportional to N.

The one escape is $K \to 0$. Only when the corrugation vanishes is the displacement limited by the chain length rather than by a restoring force, and only then does α grow faster than N.

9.2 Polarisability against chain length

Computed for a model chain — one-valence-electron atoms of core radius $r_c = 1.0\,a_0$ at spacing $a = 4.5\,a_0$, frozen partition, from the induced dipole at two field strengths well below the depinning threshold:

N	α	per atom	linearity
3	81.57	27.19	4.3%
5	137.26	27.45	5.5%
7	192.50	27.50	4.8%

Table 1: Polarisability against chain length. The last column is the spread between the two field strengths. It is the precision of the measurement rather than a departure from linearity: the fields are already at a tenth of the depinning threshold, and lowering them further puts the induced energy below the solver's own convergence noise, so 5% is the floor available here.

The polarisability per atom is constant to 1% across the three lengths, so

$$\alpha = 27.4\,N, \tag{6}$$

reproducing all three entries to better than 1% with no intercept required. The response is extensive, with $\alpha_{\text{bulk}} = 27.4\,a_0^3$ per atom.

A surface term is not resolved. At the 5% precision of these values an intercept anywhere between roughly ± 10 is admissible, so the ends of the chain cannot be shown to differ from its interior — which is consistent with the derivation, since a constraint that fixes the charge on every domain makes no distinction between an end site and a middle one, but it should be read as an absence of evidence rather than evidence of absence.

This is (5) with its coefficient supplied, and it confirms rather than tests the derivation: extensivity follows from the postulate, and what the measurement adds is the size of the coefficient and the fact that nothing anomalous happens as the chain lengthens.

One caution belongs with that coefficient, and it is not small. The partition is frozen, so e^2/K drops out of (5) entirely and what is measured is α_{intra} alone. Taking K from the corrugation at the same spacing gives $e^2/K = 73.6\,a_0^3$, nearly three times α_{bulk}. Releasing the partition would therefore roughly triple the response: the wall displacement is the *larger* part of it, and the numbers above are a lower bound on the polarisability rather than an estimate of it.

It was frozen for a reason of resolution and not of convenience. The solver moves an interface by transferring ownership of whole cells, so a wall sits at a cell boundary, while a sub-threshold field asks it to move by a fraction of one: it stays put, or jumps the full cell. A released measurement returns a staircase rather than a linear response. That is also why §8 imposes u rigidly instead of letting a field find it — an imposed displacement can be set to any value and checked exactly, since translation by one spacing must return the same energy. Releasing the partition and re-measuring α directly, which would test the factor of three above, needs a sub-cell representation of the interface and has not been done.

9.3 What the boundary treatment fixes, and what it does not

V_0 is a difference between two *wall placements*, so of every quantity reported here it is the one most exposed to how boundaries enter the kinetic energy. That exposure was measured rather than assumed. Reassembling the same converged states with the surface condition changed — the default Neumann at walls and cores, then Dirichlet at the walls, then Dirichlet at the cores — gives +379, +721 and −949 meV per electron at $a = 4.5\,a_0$.

Two readings follow, and they point opposite ways. The corrugation is *surface-dominated*: the bulk kinetic energy is nearly unchanged while V_0 moves by 1300 meV and reverses, which is direct support for the steric character argued below, since a quantity fixed at the exclusion surfaces is by definition not an electronic one. But it also means neither the magnitude nor the sign of V_0 can be read as a property of the model alone. The Dirichlet variants are a stress test rather than a rival physics — forcing the density to vanish across one cell adds a large and mesh-dependent surface term — yet the sensitivity they expose is real.

What survives is what the classification needs. $|V_0|$ is 379, 721 and 949 meV in the three treatments; it approaches zero in none of them, so K stays finite and the polarisation bounded however the surfaces are handled. The verdict is robust where the numbers are not.

9.4 Two routes, one verdict

These are different chains, as stated at the head of the section, so the agreement claimed is qualitative — that both give a bounded response and a positive curvature — and not a numerical coincidence between two determinations of one quantity.

The threshold field of §8 and the extensivity measured here are independent. One drives the partition and reports the field at which it would run; the other never applies a field large enough to move anything and reports how the bounded response scales. They rest on different assumptions — the first on a periodic energy curve and its steepest gradient, the second on the Kohn–Resta criterion and the growth of α with N — and they agree: the polarisation is bounded, the curvature is positive, and there is no DC current. That agreement is the strongest statement this article makes about the classification, because neither measurement could have produced it alone.

10 Three geometries, three regimes

The measurements above are all on one arrangement: a chain of atoms with the valence electron occupying the same region as its kernel. That is a real system — a hydrogen chain is exactly this — but it is not how conducting matter is built, and the verdict turns out to depend on the difference.

What changes, and why it should. In an ion lattice the carriers do not sit on the nuclei; the core electrons do, and the conduction charge stands outside them. Representing that needs two populations: cores held inside a radius, carriers outside it, with only the carriers free to slide. It is the one arrangement that gives a carrier a standoff *without* violating the tiling, because the inner region is genuinely owned — by the core electrons — rather than left empty. In that geometry the ion presents its net charge to the carrier from a distance, and the lattice harmonic the carrier feels is suppressed by $e^{-2\pi r/a}$.

The measurement. A finite chain has ends, and on this construction they dominate: the terminal domains bind an electron-volt harder than the interior, and a corrugation read from total energies is reading them. Displacing only the middle M carriers repairs this. The energy difference is then M times the bulk cost plus two junction costs, so differencing two window sizes cancels the junctions exactly and no terminal domain moves in either configuration:

$$\Delta E(M) = M\,\varepsilon_{\text{bulk}} + 2\varepsilon_{\text{junc}}, \qquad \varepsilon_{\text{bulk}} = \frac{\Delta E(M_2) - \Delta E(M_1)}{M_2 - M_1}. \tag{7}$$

The decomposition asserts linearity in M, which a third window tests rather than fits.

What moves is the same object throughout. Nothing in this section changes the carrier. A domain holds one unit of charge by constraint, so no charge crosses between sites; what advances is the *partition*, and it advances by its boundaries being redrawn. The free boundary is the carrier in all three arrangements below — the dislocation relation of §13, in which each carrier steps forward one spacing as the disturbance passes and charge arrives at the far end while nothing material traverses the sample. The three regimes are not three mechanisms. They are three heights of the barrier that one mechanism has to cross.

The three regimes. With the ion lattice, the standoff and the mesh held fixed:

arrangement	corrugation per carrier	regime
carrier on its own kernel	$\sim$1 eV	insulator; outside the hopping class
carrier standing off ion cores	$\approx$240 meV	inside 10–500 meV: activated hopping
carriers stiffened against modulation	$\approx$24 meV	thermal energy: unpinned

The first two are geometry: the same framework, the same interaction, asked about two different physical situations. The third is not a material but a diagnostic, and it is the informative one. Doubling the carriers' resistance to being modulated — with the ions, the standoff and the mesh unchanged — drops the corrugation by a factor of eleven. So the pinning is set by whether the carriers bunch into the wells, not by the strength of the lattice, and the question becomes what supplies that resistance.

Where the coefficient comes from, and what it is entitled to be. The $\frac{1}{2}$ in $-\frac{1}{2}\nabla^2$ is not a free choice: it is fixed by calibrating the framework against atomic spectra, hydrogen first among them. It is fixed *there*, on a bound electron in a single potential well. An extended system is a different object — as an effective mass in a solid differs from the free-electron mass without anyone regarding it as tampering — and the coefficient that governs how a carrier lattice resists being modulated by an ion lattice has no obligation to equal the one an atom's spectrum fixes. Raising it is therefore not the introduction of a parameter but a question: what stiffness does an extended arrangement actually carry? A degenerate electron gas resists density modulation because compressing it costs Fermi energy. Here a flat density is admissible on a free boundary at no cost, so the coefficient in $t(n) = C\, n^{5/3}$ is $C = 0$ exactly — §6, where it was recorded as a defect of the equation of state. It is the same missing term, and the framework is short of stiffness in two related ways: a flat density costs nothing, which is $C = 0$ and a wrong equation of state, and the penalty on a modulated density is correspondingly weak, which is what lets the carriers bunch into the wells. The measurement above probes the second. Note that the two are not interchangeable — scaling a gradient term cannot restore C, since a coefficient times zero is zero, and it is the admissibility of the flat density that the Robin condition changes. And §6 also supplies the repair — a Robin interface at $\beta a = 1.146$ restores $C = C_{\mathrm{TF}}$ exactly — so the prediction is sharp: a RealQM with the interface condition corrected should not insulate here.

What this does not deliver. A threshold that vanishes is not a conductivity. In every regime above the framework can say whether charge moves and not how fast, because the rate needs a velocity variable and a dissipation channel and a minimisation has neither. The third row says the pinning can be removed, not that a current can be computed.

Status. The corrugations above are provisional in a specific way that should be stated rather than buried. They are rigid-slide values, hence upper bounds — a kink depins lower. They come from states whose Euler–Lagrange residual is 2–3 $\times 10^{-2}$, not the 4×10^{-3} the chain reaches. The stiffened runs carry a halved timestep, so they are less relaxed than the others, and the factor of eleven rests on two coefficients rather than a curve. What is not provisional is the instrument: (7) cancels ends and junctions by construction and verifies itself against a third window.

11 The barrier is not extractable, and the control that shows it

A barrier taken as the difference of two configurations is only as good as the cancellation between their errors, and that cancellation is not something one may assume. The check is available and costs nothing conceptually: scan the carrier's position through a *full lattice period* rather than sampling two configurations, and require the scan to return to where it began. On an infinite chain $E(c{=}0)$ and $E(c{=}1)$ are the same configuration translated by one spacing and must agree exactly. Their mismatch is therefore a per-run noise floor — finite size and discretisation together — measured in the same run as the quantity it has to be compared against.

Two further benefits come with it. The barrier becomes the *amplitude* of $E(c)$, so nothing depends on having guessed where the extrema lie; and the shape of $E(c)$ is visible, so a genuine corrugation can be told from a monotone drift. Both matter, as it turns out: the maximum is generally not at the half-way point that a two-configuration measurement assumes.

Applying it is decisive. Eight scans, each nine configurations over one period, with the core radius and the lattice spacing varied:

a (a_0)	r_c	amplitude	control	ratio	
6.0	1.60	770 meV	699 meV	1.1	fails
6.0	1.60	1087	535	2.0	fails
4.5	0.80	1854	1524	1.2	fails
4.5	1.20	1735	936	1.9	fails
4.5	1.60	1900	78	24.4	passes
4.5	1.93	2597	584	4.4	passes
5.5	1.93	1949	1192	1.6	fails

Six of the eight fail their own control, and the two that pass do not survive a change of either parameter. Worse for any attempt to rescue a number from them, the passes cluster at one spacing, $a = 4.5$, across three different core radii — including one for which 4.5 is not the equilibrium. Had the control been passing because a bound chain holds the carrier, the calibrated radius at *its own* equilibrium would have passed; it fails by the widest margin in the table.

The last row is the one that settles the matter. Everywhere else in this article at least one parameter was chosen for convenience: $r_c = 1.6$ because it was to hand, $a = 6.0$ because the first scans used it. In that row both are determined — $r_c = 1.93$ from the isolated atom's ionisation energy, and $a = 5.5$ from that radius's own equilibrium, located by a five-point scan of the defect-free chain with a minimum at 5.43 a_0. *At parameters the framework does not get to choose, no barrier can be quoted.*

Why, in one line of arithmetic. The totals converge only to first order in the mesh, at about 0.91 Ha per unit h, so at $h = 0.30$ each still carries some 0.27 Ha of discretisation error — a fifth of the total — while the barrier differenced from the pair is 0.01 Ha. The quantity sought is under four per cent of the error in each quantity it is subtracted from, and would need better than ninety-six per cent cancellation to survive. It does not get it: refining the mesh alone at fixed everything else moves the value $315 \to 371 \to 552 \to 265$ meV, without converging and without monotonicity. Brute force cannot close this. Reducing the error merely to the size of the barrier needs $h \approx 0.011$, a 4200^3 grid; reaching ten per cent of it needs 42000^3.

Two artifacts, both identified. What the scans do contain is a monotone drift of 0.5–1.5 eV per site, the carrier being attracted to a chain end, superposed on a deterministic wiggle of similar size. The wiggle is grid-locking, and it can be caught in the act: at $h = 0.348$ the configurations $c = 0.125$ and $c = 0.250$ return *identical* energies to five decimals, because the cell-by-cell assignment did not flip between them. The partition is a staircase pinned to the grid, and the interface energy jumps as the staircase does. Neither artifact is noise — repeated runs of one configuration agree to all five decimals recorded — which is precisely why scatter of this kind survived so long undetected: it is exactly reproducible, and reproducibility was taken for correctness.

A caution about spacing, from the exact side. Every barrier first computed here used $a = 6.0$. On a hydrogen chain of the same geometry, full configuration interaction puts the binding at 0.8 mHa — 21 meV for six atoms, against the same atoms separated. *The chain is not bound at that spacing.* A lattice that does not hold together cannot pin a carrier, so the absence of a periodic corrugation there is the expected result and not a finding about RealQM. This invalidates the original comparison with hopping semiconductors on its own, before any question of numerical resolution arises: those materials are bound solids.

What would be needed instead. Not a finer grid. The failure is structural to the method of differencing large totals, so the routes out change what is computed. Either the barrier is obtained as a path integral of the interface force, $dE/d\lambda = \oint_\Gamma [w]\, v_n\, dS$, whose error scales with the interface rather than with the volume; or the partition is given sub-cell resolution so that E becomes a smooth function of where the walls lie instead of a rough one. Both are projects rather than adjustments, and neither was attempted here.

The control is sound, and a ring shows it. Lest the failures above be read as a defect of the specified-partition method itself, the control was checked on a geometry whose symmetry is exact. A ring of six kernels carrying six electrons, the partition rotated rigidly by ϕ in units of the spacing, must return the same energy at $\phi = 1$ as at $\phi = 0$ — the two are related by a relabelling — and must be symmetric about $\phi = \frac{1}{2}$. It is, to every digit recorded: $E(0) = E(1) = 0.56101$, $E(0.125) = E(0.875) = 0.35258$, $E(0.375) = E(0.625) = -0.21304$, with one pair disagreeing by 0.011 Ha and thereby setting the noise floor. So the method reproduces exact translations exactly, and the chain's asymmetry — its minimum displaced from $d = \frac{1}{2}$, which a reflection through a core forbids — is the finite chain's two end domains, 29% of a seven-core system, and not the partition. That ring is not otherwise usable: its angular sectors widen with radius, most of each domain lies far from its kernel, and the total energy comes out *positive* for a system six hydrogen atoms would bind at -3 Ha. It settles the control and nothing else.

Why this quantity in particular resists measurement. The construction is a Frenkel–Kontorova system, and saying so explains the difficulty rather than merely naming it. The electron domains are a chain with elastic stiffness κ supplied by Coulomb repulsion between neighbours; the kernels are a periodic substrate of amplitude set by V. Every configuration in this article is a displacement field $u(x)$ of electrons against kernels: the rigid slide is its uniform mode, and a kink is the localised profile carrying u from 0 to one spacing. In that model the width of the kink and the barrier to moving it are

$$W = \sqrt{\kappa/K}, \qquad K = V''(u_{\min}), \qquad E_{\mathrm{PN}} \sim \exp(-c\,W/a), \tag{8}$$

so *the barrier is exponentially sensitive to a ratio of two curvatures.* That is the worst possible form for direct measurement — an exponential magnifies every error in κ and K — and it is why differencing total energies was never going to work, whatever the mesh. It also indicates the route: κ and K are local curvatures, taken between configurations differing infinitesimally, where discretisation errors cancel instead of competing. Measuring those and inferring E_{PN} from (8) is a different calculation from the one attempted here, and it is left to a separate study. A caution belongs with it: (8) assumes nearest-neighbour coupling, while Coulomb is long-ranged, which changes kink profiles from exponential to power-law and shifts the transition. It is a guide to the regime, not a formula to trust to three figures.

What this says about scope. Less than this article set out to establish, and in a different direction. No activation energy is available, so the construction is neither placed in the hopping-semiconductor class nor excluded from it by measurement. The scope claim of §5 stands or falls on the structural argument instead — non-overlap, a vanishing transfer integral, and the absence of any electron equation of motion — which is where it should have rested from the beginning, since those follow from the postulate and need no grid at all.

Earlier figures, and why they are not quoted. A series of barrier scans was made before the difficulty above was understood, all of them by the free-boundary route: a compression test on a ring, a doping series at two concentrations, a lattice barrier between interstitial pockets of order 5 eV, and a localised interface defect costing 179 meV independently of chain length. Every one specified or evolved a partition that no symmetry fixes, so each is a bound of its stated order rather than a determination, and all are superseded by the barrier above. What survives from them is the direction of the density trend, which the symmetry-fixed measurement confirms.

What is actually transported. The carrier in this framework is not an electron. An electron here is not a particle but a unit of charge density on a domain, and what exists is the *partition* of space; dynamics is the evolution of that partition, and transport is the partition pattern travelling. What moves along the chain is the **free boundary** — a locus where two densities balance — and nothing material traverses it at all.

The quantitative form of this is the useful one. As a kink passes, each electron advances by exactly one site spacing and then stops, while the kink advances the entire length of the chain. The carrier's displacement and its constituents' displacement are decoupled, and their ratio is the chain length. This is the relation between a dislocation and the slip it produces: the dislocation crosses the crystal, no atom moves far. Charge is genuinely transported and real charge arrives at the far end; it is the *carrier* that is immaterial, in the sense that a wave is immaterial while the water is not.

That diagnosis also accounts for the failures recorded earlier in this study rather than excusing them. Each of the unsuccessful constructions imposed a carrier that had to *translate*, moving bodily through occupied space, and returned costs of tens of electron-volts or else collapsed. Those were the price of forbidding the only mechanism the framework has. The solver states as much directly: its free boundary evolves by the density balance $(\rho_i - \rho_j)/(\rho_i + \rho_j)$, flipping ownership cell by cell. Boundary motion is the only motion it performs. The corona result is the same statement seen from the favourable side — an added electron spread into a uniform shell at no cost, which is a boundary redistributing freely.

This raises the stake of the kink measurement. If transport works here it works *because* of the free boundary, which is the single feature separating this framework from the standard theory; and if the boundary cannot carry charge cheaply, the difficulty is not in a detail of the construction but in the framework's defining object.

A repair that is not available, and one that is. An obvious response to the barrier is to let valence electrons share a territory, which removes it by construction. The $n^{5/3}$ scaling rules that out: sharing destroys the extensivity the partition supplies, and a metal would lose its kinetic energy density by fifteen orders of magnitude.

What the framework then is. Non-overlapping sites, occupation 0, 1 or 2, an on-site repulsion U, and transfer between neighbours: that is the **Hubbard model**. Which locates RealQM exactly. As formulated — one unit of charge per domain, always — it is the Hubbard model in the limit $U \to \infty$, where double occupancy is forbidden outright and every site is singly occupied. That limit is a Mott insulator at every density, which is precisely what the earlier barrier scans found by direct computation, and it explains the density trend as well: no finite U can be beaten by a bandwidth that does not exist.

The repair is correspondingly minimal. Keep the partition, keep the geometry, keep the Coulomb energetics, and allow occupation to take the value two at a computed cost. The framework then sits where the standard treatment of strongly correlated matter sits — the Hubbard model is the working description of Mott insulators, of the cuprates, and of the materials for which band theory fails — with the difference that its U and its lattice are derived rather than supplied.

We do not carry this out here. What is established is where the obstruction lies: not in the interface condition, not in the geometry, but in the rule of one unit of charge per domain. That rule is also the source of one of the framework's results, charge quantization as a consequence of indivisibility rather than an assumption. The tension is worth stating plainly: *the postulate that explains why charge comes in units is the postulate that forbids a metal.* Matter has it both ways, quantizing the particles while leaving the amplitude on each site continuous, and a partition of real densities has no evident means to do the same.

12 The carrier itself, against exact diagonalisation

Everything above is internal to the framework. One external check is worth keeping, and the sharpest form of it does not compare energies at all. Take the same object this article hops — n protons, $n-1$ electrons, nuclei clamped — and diagonalise it exactly in a minimal basis. The Loewdin site populations then say directly whether the carrier is localised:

a (a_0)	E_{corr} (eV)	$4t$ (eV)	max hole/site	
3.0	-6.43	11.11	0.223	delocalised
4.0	-11.97	4.65	0.215	delocalised
4.5	-14.64	2.96	0.226	delocalised
6.0	-19.86	0.71	0.253	delocalised

A hole spread evenly over seven sites reads 0.143; one sitting on a single site reads 1.000. At every spacing the ground state puts it on four or five. *There is no localised carrier, and therefore no barrier for one to cross.* Not because correlation is weak — E_{corr} reaches 19.9 eV — but because strong correlation and localisation are different things: a one-dimensional Mott insulator doped with a single hole conducts, the holon propagating freely.

The disagreement is therefore about the *state* and not about a number, which is the harder kind to explain away. Two qualifications keep it honest: a minimal basis is exact diagonalisation within a deliberately crude span — STO-6G misses the hydrogen atom by six per cent before any chemistry begins — and larger bases did not converge for the stretched chains, so no exact reference exists for these configurations and nothing here declares the pinning wrong for them. What it does establish is the direction of the error. The construction over-localises, by construction, being $U \to \infty$: it fails as §5 said it would rather than arbitrarily, and where the mechanism of this article would actually be tested — a doped Mott insulator, an organic semiconductor, a polaronic oxide, where on-site repulsion and not band structure sets the gap — no comparison is made here.

Two controls added after the fact, and what each is blind to. The first is free and should have been applied from the start: *domains related by a symmetry of the configuration must have equal eigenvalues.* Each domain's stationarity condition carries an ε_i, and on a chain where every site has the same environment they must coincide. A total energy is one number and can look reasonable while the state beneath it is inhomogeneous; the eigenvalues expose that directly. Applied to the coaxial geometry of the follow-on work it showed terminal domains binding an electron-volt harder than interior ones — which identified a corrugation as an end artifact rather than a registry effect — and, on longer chains, a smooth bowl across the whole chain that distinguishes a slowly converging one-dimensional Coulomb sum from a local end effect. It is blind to exactly what the mesh ladder and the translation control are blind to: an error respecting the lattice symmetry passes all three, which is how the exclusion rule above survived them.

The second is a way to measure a bulk quantity on a chain that has ends. Displace only the middle M carriers by half a spacing: the energy difference is M times the bulk cost plus two junction costs, so differencing two window sizes cancels the junctions exactly and no terminal domain moves in either configuration. The decomposition asserts linearity in M, which a third window tests rather than fits. It is the instrument the rigid slide of a finite chain should have used from the beginning.

A note on what was tested numerically

The plasma frequency $\omega_p^2 = 4\pi n$, the screening relation, the Thomas–Fermi comparison and the vanishing of C are analytic. The alkali lattice constants are a radial Wigner–Seitz calculation with the stated pseudopotential and correlation functional, run for five alkalis; the standard column reproducing the series is what makes the comparison meaningful, and its own 44% mean error in the bulk modulus sets the accuracy available at this level of model. Three attempts are reported as null rather than omitted. The interface parameter β was tested on helium at three meshes at fixed imaginary time, giving shifts of $+0.151$, -0.018 and -0.184 Ha as the mesh refined — swinging through zero and growing, with absolute energies moving away from the exact -2.9037. The cause is the geometry rather than the arithmetic: in helium the two domains meet at a plane *through the nucleus*, so interface error and Coulomb-cusp error are superimposed and cannot be separated on a uniform grid. The measurement wants H_2, where the domains meet at the midplane *between* the nuclei in a smooth region. An interface-condition test on helium was not converged, giving a sensitivity that reversed sign under mesh refinement, and no conclusion is drawn from it. Two chain calculations — five hydrogens with an excess electron, and five hydrogen kernels with one electron missing — returned no transport, but for a reason established by reading the constructions rather than by the runs: a carrier without a kernel feels no lattice force, and a bare kernel receives no domain for a vacancy to occupy. Both are recorded because the null results located those obstructions. A caution on the numerics belongs with them. Several attempts returned energies below the physical floor for their particle content, but these track the boundary speed and the seeding rather than the configuration: with the interface driven faster than the densities can follow, or with a carrier seeded directly onto an occupied kernel, the relaxation runs away, while the same configurations at a quarter of that speed settled to physical values. A two-domain shell on one nucleus is in fact stable under the density-balance rule — compressing the inner domain raises its density at the interface and lowers the outer's, which pushes the boundary back out — and both domains carry real kinetic energy there, since the nucleus shapes their profiles. The vanishing of C is a statement about a *uniform* density and does not transfer to a shaped one. Scripts and pages accompany the source.

13 Conclusion

The question was whether a construction of non-overlapping unit charge densities, meeting at free boundaries, conducts. It comes in the two tiers of §4, and the formulation answers one of them.

The **classification** it settles, and settles from a static calculation, which is what Kohn's result licenses: the polarisation is bounded, the curvature positive, the threshold field beyond dielectric breakdown, and α extensive in the number of atoms. This is an insulator, and not by a near miss. What cannot be reached is the **rate** — a conductivity, or the activation energy that would set one — and the reasons for that are worth more than the number would have been.

The carrier is a free boundary, and that identification holds. An electron here is a unit of charge density on a domain; what exists is the partition of space. Transport is therefore the partition pattern travelling, in the relation a dislocation bears to slip: charge arrives at the far end while nothing material crosses the sample. That picture accounts for every null result reported above — each had imposed a carrier obliged to translate bodily through occupied space — and it is why the one transport calculation this framework performs successfully is Grotthuss conduction, where the carrier is a proton and the framework does have an equation of motion for it.

The picture has a definite limit, and it should be stated with the claim rather than after it. Defect-mediated transport is real — dislocation glide, charge-density-wave sliding and ionic conduction all proceed that way — but it is not how a metal conducts. Residual resistivity *falls* as defects are removed, so a carrier that must migrate as a defect predicts the opposite of what purity does. The free boundary is therefore the mechanism this construction is restricted to, not the carrier of ordinary conduction, and its prominence here is a fact about the formalism rather than a discovery about charge transport.

Costing that motion defeats the method, and the control is what shows it. A barrier of 0.01 Ha differenced from totals of 1.3 Ha carrying 0.27 Ha of first-order discretisation error requires a cancellation it does not get, and brute force cannot supply it: reaching ten per cent of the barrier needs a 42000^3 grid. Scanning a full lattice period instead of sampling two configurations supplies the diagnosis for free, since translation by one spacing must return the same energy. Six of eight such scans fail that control, the two that pass survive no change of parameter, and the only scan at parameters the framework does not choose fails by the widest margin. What the scans do contain is a drift of the carrier toward a chain end and grid-locking of a staircase interface — both of order an electron-volt, both exactly reproducible, which is why they were mistaken for signal for as long as they were.

But the question does not require that quantity. Asked as a threshold field rather than as a barrier — and on a cyclic chain, where translation by one spacing is an exact relabelling and the control holds by construction — it becomes the maximum gradient of a measured periodic curve, and §8 gives $7\text{–}9\times10^9$ V/m, twelve orders above the field at which copper carries an ampere and two to three orders above dielectric breakdown. The same curve yields a pinning potential of 812 meV per electron and a positive curvature, hence bounded polarisation and no DC current. The verdict the article set out to reach is therefore available after all; what was never available was the number it first tried to reach it with.

Beneath that lies a structural obstacle no refinement touches. Non-overlap makes the transfer integral vanish identically, so band conduction is absent in kind: there are no extended states to carry it. And in this scheme nuclei move under classical equations of motion while electrons relax to a static minimum — so there is no electron dynamics, and hence no object in the theory corresponding to a current. *A minimisation has no time in it.* Every attempt above infers transport from an energy landscape, which is a proxy, and it is the proxy that breaks. The missing ingredient is not a better barrier calculation but an equation of motion, and it exists elsewhere in the programme: the time-dependent form in which a real density carries a complex current is where a conduction calculation belongs — and it supplies the current, not its rate, which needs the dissipation no framework derives unaided.

What the framework is, and is not. Single occupancy with long-range Coulomb is *not* the Hubbard atomic limit, whose degeneracy over carrier position would give a barrier of exactly zero; the Coulomb interaction between extended densities breaks that degeneracy, placing the construction closer to an extended Hubbard model where charge ordering is genuine physics. Within bound matter — structure, binding, geometry — it works, and the transport regimes it reaches are those with nuclear carriers. An electronic transport *rate* is outside, and against exact diagonalisation on the same chains the reason is visible directly: the exact carrier occupies four or five sites where this construction admits only one.

On the system tested. Every measurement reported here is on a regular chain of atoms carrying one valence electron each. The extension to a regular two- or three-dimensional lattice needs nothing new — the machinery is three-dimensional already and only the kernel positions change — but it has not been made, and a partition with more neighbours to slide against is where the corrugation would next be tested.

What the time-dependent form would have to supply. That the framework is static here is a fact about this article, not about RealQM, and the route out is specific enough to state. What has to be supplied is a velocity, and it need not arrive as a complex phase: a real density carrying a real velocity field is Madelung's form of the same equation [4], and in this framework the natural variables are the densities together with the velocities of the boundaries that bound them, with $J = \rho v$ and continuity. Either way a current exists as an object, and the first obstacle goes. It does not remove the second, since a Hamiltonian dynamics under a constant field rings rather than settles, exactly as the $3N$ equation does, and a steady current needs dissipation. But the dissipation is already in the scheme and need not be imported: the nuclei move under classical equations of motion, so energy can leave the electrons into lattice motion, and the electron–lattice channel that §4 found every treatment inserting by hand — a collision term, a partial trace, thermalising reservoirs — would here be present in the equations being solved. A resistivity obtained that way would have τ as an output. That is the strongest form the claim of this programme could take on transport, and nothing in this article establishes it. What time-dependence would *not* change is the exclusion: a phase does not make domains overlap, so there is still no transfer integral and no extended states, and the target remains activated transport with localised carriers.

The verdict depends on the arrangement, which is the article's main change of view. Measured on a chain where the carrier occupies its own kernel's region, this framework insulates and falls outside even the activated-hopping class it claims. Measured on a wire — cores inside, carriers standing off them, which is how conducting matter is built — the corrugation drops into that class. And given the stiffness against density modulation that a degenerate gas has, it drops to thermal energy and the pinning goes. So the insulating verdict of §8 is a verdict about a geometry, not about the framework, and the framework's real obstruction is the one §6 already identified for the equation of state: the free interface leaves the partition without stiffness. That the same missing term should produce both a wrong compressibility and an insulator is the most useful thing this article has to report, and it comes with its own repair — the Robin interface at $\beta a = 1.146$ — which makes it a prediction rather than a diagnosis.

The result is therefore a boundary, not a barrier. A method with a mapped edge is more usable than one claimed to be universal, and the edge is worth stating exactly rather than sweepingly, since its four parts are not of the same kind. *Metallic conduction* is excluded structurally: non-overlap leaves no extended states and no transfer integral, and no refinement produces them.

A *conductivity* is unavailable in the static form, which has no phase, hence no current as an object and no clock to give one a rate. *Nuclear* transport is inside, protonic conduction being the one transport calculation that works. And electronic transport by boundary motion is *described rather than forbidden*: the partition slides, the free boundary carries the charge, and what this article reports is not the absence of that mechanism but its pinning — on one chain, in the rigid mode, on a clamped lattice, with two upper bounds that both push the number down. A pinned mechanism is a described one. Everything this article originally set out to measure — a carrier barrier, a hopping rate — lay on the far side of the first two.

References

[1] C. Johnson, *Real Quantum Mechanics*, `https://realqm.com`.

[2] C. Johnson, *Real Quantum Mechanics: a new view of the atom*, book manuscript.

[3] C. Johnson, *RealNucleus: nucleons as charge densities*, submitted.

[4] E. Madelung, *Quantentheorie in hydrodynamischer Form*, Z. Phys. **40** (1927) 322.